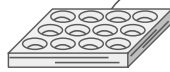
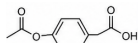


Input data

Small Molecules & Toxicity Assays

small-molecule



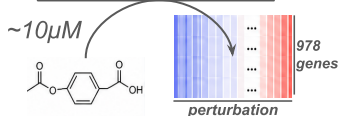
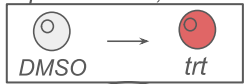
→ 1 (toxic)
→ 0 (non-toxic)

Assay

Twelve toxicity endpoints
(Tox21 Data Challenge)

Perturbation Transcriptomics

same plate,
exposure time, cell line



small-molecule
(at a fixed dose)

Four cell lines
(LINCS Level 3)

Stage 1: Frozen Molecular Prediction

Molecular structure model

01011

Molecular Fingerprint + Machine Learning
MACCS, RDKit, and Morgan (ECFP4)



Molecular Graph Models
GCN, GAT, GraphSAGE, GIN, and Chemprop

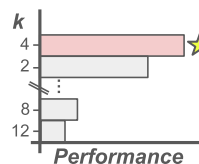


Molecular Language Models
ChemBERTa

Endpoint-specific models on twelve configurations ($k = 12$)

Ensemble selection

Selection from top- k models



Equal-weight probability mean ensemble

Frozen molecular reference model

Frozen logit as fixed offset for Stage 2

$$\eta^{\text{chem}} = \text{logit} \left(\frac{1}{k^*} \sum_{m \in \mathcal{M}(k^*)} p_m^{\text{chem}} \right)$$

molecular logit

Ensemble prediction

η^{chem} Frozen
Molecular structure-based prediction

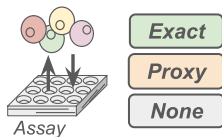
Stage 2: Transcriptome-complemented Prediction

Context-matched cohort

Cell Line



Cellular Background



Exposure Time



Transcriptomic residual model

Ridge logistic regression via context-matched gene-expression profiles

$$\eta^{\text{chem+bio}} = \underbrace{\eta^{\text{chem}}}_{\text{frozen molecular offset}} + \underbrace{\alpha^{\text{chem+bio}}}_{\text{calibration}} + \underbrace{\tilde{\mathbf{x}}^T \boldsymbol{\beta}}_{\text{transcriptomic term}}$$

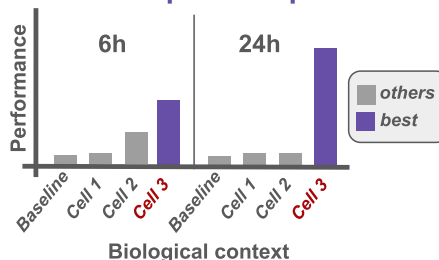
transcriptome-complemented logit

$\eta^{\text{chem+bio}}$

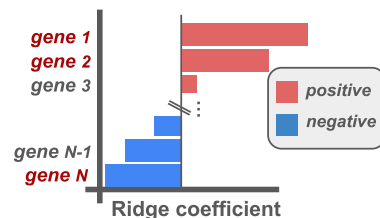
Biological correction beyond the molecular model

Outputs & Post-hoc Analyses

Context-dependent prediction



Gene-level analysis



Compound-level analysis

Toxicity Pathway

